

220B Discussion 6

Problem 1: “The best equivariant estimator” depends on the loss

Let $X_1, \dots, X_n$ be iid $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 > 0$ unknown, and let $S = \sum_{i=1}^n X_i^2$. Throughout, $W := S/\sigma^2 \sim \chi_n^2$, so $\mathbb{E}[W] = n$ and $\mathbb{E}[W^2] = n(n+2)$.

The problem is invariant under the scale group $\mathcal{G} = \{g_b : g_b(x) = bx, b > 0\}$, with induced actions $\bar{g}_b(\sigma^2) = b^2\sigma^2$ and $\tilde{g}_b(a) = b^2a$.

(a) Show that any equivariant decision rule based on S has the form $\delta(S) = cS$ for some constant $c > 0$.

Now consider two scale-invariant loss functions:

$$L_1(\sigma^2, a) = \left(\frac{a}{\sigma^2} - 1\right)^2, \quad L_2(\sigma^2, a) = \frac{a}{\sigma^2} - \log\left(\frac{a}{\sigma^2}\right) - 1.$$

(L_2 is **Stein's loss**, also called the entropy loss. Both are invariant since they depend on a and σ^2 only through a/σ^2 .)

(b) Under L_1 , compute $R(\sigma^2, cS)$ in closed form. Verify the risk does not depend on σ^2 . Minimize over $c > 0$ to find the best equivariant estimator

$$\delta_1^*(S) = \frac{S}{n+2}.$$

Problem 1 (continued)

(c) Under L_2 , compute $R(\sigma^2, cS)$ in closed form. (You will need $\mathbb{E}[\log W]$, but this term is a constant in c and drops out of the minimization, so you do not need to evaluate it.) Minimize over $c > 0$ to find the best equivariant estimator

$$\delta_2^*(S) = \frac{S}{n} = \text{the MLE.}$$

(d) The model is $\mathcal{N}(0, \sigma^2)$, the group is the scale group, the sufficient statistic is S , the equivariant class is $\{cS : c > 0\}$. None of this changed between (b) and (c). Yet the best equivariant estimator is $S/(n+2)$ under L_1 and S/n under L_2 .